

Labelled unemployed: Neighbourhood composition and the enforcement of employment norms

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Abstract

This article illuminates the contexts in which neighbourhood unemployment serves as a mechanism to explain the unemployed's perception of being negatively stereotyped, also known as stigma consciousness. The article relies on the labelling approach and social contagion models to derive hypotheses about the effect of informal societal control and the scope of the employment norm. After combining survey data with administrative 1 × 1 km grid cell data from Germany, multilevel models reveal a U-shaped association: the neighbourhood's unemployment negatively affects the individuals' stigma consciousness up to an unemployment share of 10% and positively affects the individual's stigma consciousness when the neighbourhood share exceeds 30%. Especially the unemployed in high-inequality neighbourhoods and low-unemployment counties show the highest associations between neighbourhood unemployment and stigma consciousness. Beyond highlighting the importance of local norms and how they shape individuals' perceptions in general, the article sheds light on how employment norms operate on different spatial levels.

Keywords

labelling approach, neighbourhoods, social contagion, unemployment

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Introduction

Why do some unemployed experience intense social stigma for the same behaviour that others carry without reproach? While society treats paid work as a central moral and social standard (Rogers-Dillon, 1995), not all unemployed individuals perceive – or experience – the same degree of social disapproval (Lang and Gross, 2019). This heterogeneity raises a fundamental sociological question: How do local social environments shape the enforcement of employment norms and why do some communities enforce them more rigorously than others?

Two complementary research traditions speak to this question. The labelling approach highlights that stigma emerges through processes of social evaluation in concrete interactions (Lang, 2022; Lang and Gross, 2019). Neighbourhood effects research shows that local contexts influence the salience and enforcement of norms (Sampson, 2012; Vandecasteele and Fasang, 2021). Yet these literatures rarely intersect, leaving a gap in understanding how the structural and spatial dimensions of social life shape the enforcement of work-related norms.

Addressing this gap, the article asks the question: How does the prevalence of norm deviation affect the feeling of stigmatisation? More specifically, how does the local neighbourhood unemployment affect the stigma consciousness of the unemployed? Under which conditions are neighbourhood unemployment and the unemployed's feeling of stigmatisation interrelated?

In theoretical terms, the article integrates the labelling approach (Becker, 1973; Erikson, 1962; Keckeisen, 1974; Kitsuse, 1962; Lemert, 1951; Tannenbaum, 1938) with social contagion theory (Crane, 1991; Mayer and Jencks, 1989) to conceptualise stigma (consciousness) as a form of informal social control. The proposed theoretical model argues that neighbourhood composition shapes the visibility and evaluation of unemployment, producing non-linear effects: the unemployed's stigma consciousness decreases as neighbourhood unemployment rises up to moderate levels, but increases again under conditions of high unemployment, particularly in local coexistence with socio-economically advantaged individuals or when entire neighbourhoods are embedded in more affluent regions.

Empirically, the article links survey data from the German Panel Study Labour Market and Social Security (PASS), which directly measures stigma consciousness among unemployed individuals, to fine-grained administrative 1×1 km grid cell data. The combination of survey and administrative context data allows a detailed examination of how local social environments structure perceptions of stigma.

In most European countries, welfare states have shifted toward 'activation' policies that emphasise individual responsibility and push citizens into employment (e.g. Trickey and Walker, 2001; Weishaupt, 2010). Germany provides a particularly instructive case for exploring these dynamics. Both an informal (Chadi, 2014; Kronauer et al., 1993; Nonnenmacher, 2009) and formal¹ employment norm exist (Lang, 2022). In Germany, employment is a highly visible and morally charged standard of social worth. At the same time, it functions as a powerful boundary between social groups (Barlösius, 2002). In trying to break the link between social worth and employment, a policy reform in 2023 renamed 'unemployment benefits' (German 'Arbeitslosengeld II') to 'civil benefits'

(German 'Bürgergeld'). Besides this institutional salience of the employment norm, Germany exhibits substantial spatial variation in unemployment and inequality, from affluent urban centres to structurally disadvantaged post-industrial regions. This spatial variation allows us to examine fundamental sociological processes of norm enforcement and stigma in ways that generalise beyond the German context.

The results reveal three consistent patterns. First, the relationship between neighbourhood unemployment and stigma consciousness is U-shaped, with the lowest stigma consciousness in neighbourhoods with moderate unemployment and rising stigma consciousness in contexts of high unemployment. Second, larger spatial contexts moderate the effects of local unemployment: stigma consciousness is amplified when high-unemployment neighbourhoods are embedded in low-unemployment counties. Third, local coexistence with affluent residents intensifies the link between neighbourhood unemployment and stigma consciousness by making social boundaries between employed and unemployed residents more visible.

These findings contribute to work and employment research in three ways. First, they demonstrate how combining the labelling approach with social contagion theory clarifies the context-dependent enforcement of employment norms. Second, they provide the first direct evidence linking neighbourhood composition to stigma consciousness using representative survey data integrated with administrative georeferenced data. Third, they show that the social enforcement of employment norms is structured not only by individual perceptions but also by the spatial and social organisation of communities, offering a generalisable account of how social context shapes the lived experience of norm violation. This gives rise to a whole range of new research approaches, for example, concerning decision-making processes in the labour market in the face of impending unemployment and the associated threat of stigmatisation.

Spatial contexts of norm setting and enforcement

Employment norms

In general, several social norms on employment exist. The labour market participation of women differs depending on the prevalence of gendered work norms (Han et al., 2024; Sprengholz et al., 2022). Employees stay late in the office to conform to ideal worker norms (Lott and Klenner, 2018; Williams, 2001). But also the absence of employment, that is, unemployment, is connected to social norms. Social scientists have long discussed the concept of welfare stigma as a negative social response to receiving public assistance (Eberl et al., 2023; Horan and Austin, 1974; Moffitt, 1983; Pohlan, 2024; Rogers-Dillon, 1995). Even individuals who leave the labour market for health reasons, and are therefore perceived as deserving, use legitimising narratives to cope with stigmatisation (Hiebert et al., 2024). The reasons for amplified feelings of stigmatisation and coping mechanisms are linked to social responses to norm-deviant behaviour, such as social distancing or moral condemnation (Bicchieri, 2005; Gurr and Jungbauer-Gans, 2017; Gurr and Lang, 2018). These studies show how employment norms, once violated, can trigger emotional distress – a clear sign of how norms shape lived experiences.

To assess differences in the feeling of being the target of such social responses, the concept of stigma consciousness becomes central. It captures whether individuals expect to be judged or excluded because of their unemployment (Gurr and Jungbauer-Gans, 2013). Where the employment norm and the associated informal social control are strong, the unemployed are more likely to be conscious of their stigma. In contrast, in contexts with weaker social control, people may not internalise such stigma. Focusing on stigma *consciousness* rather than stigmatisation accounts for how individuals' own perceptions and emotions heavily influence the consequences of stigmatisation. This is, among others, a central point of criticism of research on stigmatisation as formulated by Pinel (1999), Link and Phelan (2001), or Egdell and Beck (2020). Pinel (1999) argues that targets of stigmatisation do not perceive these attributions in the same way and stresses the importance of focusing on stigma consciousness to better account for these perceptions.

The neighbourhood context

While several studies have argued and empirically shown how social norms emerge and affect individuals' outcomes differently depending on their own characteristics (Lang and Gross, 2019), less is known about the contexts in which employment norms evolve and intensify. This article adopts a constructivist perspective on norms (Bicchieri, 2005; Horne and Mollborn, 2020) and argues that employment norms are context-dependent. The constructivist theory of norms states that the existence of a social norm depends on a sufficient number of people believing in its existence and following the norm (Bicchieri, 2005). In this view, both the presence of individuals who deviate from the employment norm as well as those who uphold and enforce it are central to understanding how stigma consciousness among the unemployed emerges.

To derive how neighbourhood contexts relate to the effectiveness of social norms, the article employs social contagion theory and the labelling approach. Focusing on the diffusion of local norms, contagion models suggest that people learn what is acceptable by observing and interacting with neighbours (Mayer and Jencks, 1989; Sampson, 2012). In neighbourhoods with many unemployed residents, individuals may perceive unemployment as normal, also beyond their local context (Bisgaard et al., 2016). This does not mean the employment norm disappears entirely, but that the enforcement weakens because fewer people actively reinforce it. The more frequently unemployed individuals interact with peers in similar situations, the less likely they are to feel stigmatised. Social contagion theory thus predicts that unemployed people in high-unemployment neighbourhoods experience lower stigma consciousness.

A large body of literature has shown that neighbourhood (un)employment is connected to (fewer) material resources, (low-quality) networks (e.g., Ioannides and Loury, 2004; Jahn and Neugart, 2020; Smith, 2005), and (poor) role models (e.g., Ostermann and Bähr, 2025), all leading to (adverse) labour market chances. For social contagion mechanisms, some existing international research finds positive effects of local unemployment on individuals' well-being and their probability to receive welfare benefits (Celhay et al., 2025; Clark, 2003; Howley and Knight, 2022). Studies from Germany provide more ambiguous evidence for a social contagion effect on individuals' life satisfaction (Clark et al., 2010; Oesch and Lipps, 2013). These ambiguous findings may be

due to missing direct measures of perceived stigmatisation or relatively broad levels of measuring local (un)employment norms.

The labelling approach complements the considerations derived from social contagion theory by emphasising the interactions between deviant individuals and norm-enforcing others. Despite being written much earlier than Bicchieri (2005), the labelling approach also argues that deviance does not lie in an act or a status itself, but in the social reaction to it (Tannenbaum, 1938) – and thus takes a similarly constructivist perspective as Bicchieri (2005). Therefore, stigma arises when norm deviation is observed and negatively defined as such by others. For the unemployed to feel stigmatised, there must not only be a deviation from the norm (e.g., being unemployed) but also an audience that recognises and negatively evaluates being unemployed as a deviation from the local norm. In the absence of norm-conforming and potentially stigmatising others, deviant behaviour may go unchallenged or unnoticed.

While the labelling approach is still applied primarily within criminology and research on deviant or criminal behaviour (Lee, 2024; Wingert et al., 2025; Xu et al., 2021), research linking this approach to labour market outcomes, for example in transition to work (Cerdeira-Jara et al., 2024) or within employment (De La Haye et al., 2023), exists. Still, the labelling approach can be applied in different contexts of norms and norm deviation as well (Lang, 2022; Lang and Gross, 2019).

From a labelling approach perspective, two mechanisms drive the link between neighbourhood unemployment and stigma consciousness of the unemployed: the internalisation of a deviant social identity and informal social sanctions (Lang and Gross, 2019). With higher neighbourhood unemployment, the unemployed are more likely to perceive themselves no longer as deviant individuals but as part of a deviant group. Internalising this perception as part of their social identity can reduce stigma consciousness. However, increasing visibility of deviance may counter the social identity mechanism, where visibility leads to stronger informal sanctioning, which reinforces the unemployed's stigma consciousness. As these theoretical mechanisms are difficult to test empirically, this article focuses on the overarching question of the overall effect of neighbourhood unemployment.

Combining social contagion theory and the labelling approach leads to a hypothesis on a U-shaped relationship: stigma consciousness may decline as more individuals in their environment deviate from the employment norm, thereby reducing the perceived deviance of their status. However, if the group of unemployed is large enough to become visible as a distinct social group, stigma consciousness may intensify rather than diminish.

H1: *The stigma consciousness of the unemployed is higher in neighbourhoods with low and very high shares of unemployment compared to neighbourhoods with medium shares of unemployment.*

Spatial moderation of employment norms

The article proposes two spatial moderators staging this trade-off. The first moderator follows a nesting² structure, where the group of norm-deviant individuals is embedded in a larger environment with intact, reinforced employment norms. Previous literature on

norms has shown that regional contexts play an important role in explaining variance in gendered employment norms in Europe (Cascella et al., 2024) and Germany in particular (Sprengholz et al., 2022; Zoch, 2021). It is less clear, however, whether these regional associations are truly regional or rather aggregated products of more local interaction-based mechanisms, such as informal social control.

Following Jones et al. (2018), the theoretical model of nested spatial units predicts distinct individual-level outcomes for specific combinations of county-neighbourhood unemployment rates, rather than merely controlling for larger spatial environments. In cities with generally low unemployment rates, neighbourhoods with many unemployed individuals can signal a shared stigmatised identity (Becker, 1973; Keckeisen, 1974) and foster ‘territorial stigmatization’ (Wacquant, 2008). Wacquant et al. (2014) describe territorial stigmatisation as a public discourse that labels, among others, poor urban neighbourhoods as ‘threats to [. . .] the fabric of the nation’ (p. 1273) and ‘vectors of social disintegration’ (p. 1274). Individuals from such stigmatised neighbourhoods are more likely to experience or perceive stigmatisation in other spheres of life; directly, like on job search (Smith, 2005) and economic transactions (Besbris et al., 2015) or indirectly via the local news (Schwarze, 2022).

However, territorial stigmatisation attaches negative symbolic meanings not only to individuals but also to the places they inhabit (Wacquant, 2008). As a result, unemployed individuals living in highly stigmatised neighbourhoods may experience unemployment not merely as an individual status but as part of a collectively discredited social identity. This additional layer of stigmatisation should reinforce awareness of one’s own deviant status rather than normalising it. Hence, nested norm structures, for example, high-unemployment neighbourhoods in low-unemployment broader spatial units, should amplify the ‘positive’, that is, increasing effect of high neighbourhood unemployment on the unemployed’s stigma consciousness.

The second spatial moderator concerns the local coexistence of socio-economically advantaged and disadvantaged residents within neighbourhoods. In contexts where unemployed and affluent individuals live in proximity, social comparison processes may be more salient and the contrast between norm conforming and non-conforming behaviour becomes more visible (Sharkey and Faber, 2014: 563). Hence, we conceptualise coexistence of different groups not as diversity per se, but as the spatial proximity of groups occupying different positions relative to the employment norm. Income-based inequality proxies these different positions beyond current unemployment episodes. Also, income differences manifest in different types of clothing, house types, and cars, which can serve as social cues for exclusion practices of affluent coresidents against less affluent residents (Paulsen and Stuber, 2022). This is in line with relative deprivation theory (Runciman, 1966), stating that individuals may more acutely perceive their deviation from dominant expectations when exposed to better-off individuals.

In essentially being ingroup-outgroup behaviour, strong norm enforcement and stigmatisation only occur when both groups are substantial in size but distinct from each other (Galster, 2014). If the number of deviant – potentially stigmatised – persons is small, no enforcement of a set norm is necessary. Conversely, if the number of stigmatising persons is small, the norm enforcement is ineffective. Strong norm enforcement should occur when not only a sufficient local number of potentially stigmatised persons

but also stigmatising persons are present, that is when a neighbourhood's socio-economic heterogeneity is high. These considerations lead to two additional hypotheses:

H2: *The stigma consciousness of the unemployed living in low-unemployment counties but high-unemployment neighbourhoods is higher than that of otherwise similar unemployed living in high-unemployment counties (nesting hypothesis).*

H3: *The stigma consciousness of the unemployed living in socio-economically heterogeneous but high-unemployment neighbourhoods is higher compared to otherwise similar unemployed who live in more homogeneous neighbourhoods (socio-economic coexistence hypothesis).*

Data and methods

Data sets

The article tested the hypotheses by combining rich survey data from one wave of the PASS with lagged administrative 1×1 km grid cell neighbourhood data.

In general, the PASS is a panel data set and comprises two separate initial subsamples. The first subsample was drawn from all households residing in Germany with at least one individual receiving social benefits at one of the seven sampling dates, while the second subsample stems from the full population of all private households in Germany (Trappmann et al., 2019). The article used only the subsample of unemployed respondents from 2013, because the item battery on stigma consciousness is only part of the PASS for the unemployed in wave 7. The analysis sample comprised 1841 unemployed individuals being older than 16 years and residing in Germany at the time of the interview. In the analysis sample, women were slightly over-represented, respondents are on average 45 years old, and had children in 27% of all cases (Table 1). Roughly 50% of all respondents lived in urban counties.

To approximate the theoretical concept of the local deviation from employment norms, the article uses unemployment rates on a 1×1 km grid-cell level. While grid-cell data insufficiently capture socially meaningful neighbourhood boundaries (Ostermann, 2026), grids measure spatial proximity. In turn, spatial proximity is an important factor for local communities (Preciado et al., 2012). The employed grid-cell data, the GridAB, is a trend data set from the IAB containing aggregated register data from the Federal Employment Agency for the whole of Germany and the years 2000 to 2017. The underlying register data provide highly reliable and detailed information on all employed, the registered unemployed, and individuals being targets of active labour market policies. The full population on 30 June of each year was aggregated to 1×1 km grid cells, leading to 244 residents per grid cell on average (for an example, see Figure 1 and for detailed information about the grid cell data, see Ostermann et al. (2022)). Uninhabited grid cells are not in the data. As the georeferenced data are highly sensitive, neighbourhoods with fewer than ten inhabitants are also not included. As the address information from the survey served as the spatial identifier, 99.51% of the analysis sample could be merged with valid information from 1179 grid cells in 2012. On average, 1.6 respondents reside

Table 1. Sample descriptives.

	Analysis sample Mean/sd
Individual characteristics	
Stigma consciousness	38.35 13.55
Female, ref. male	0.45 0.50
Has children, ref. childless	0.27 0.44
Care obligations, ref. no care obligations	0.09 0.28
Age	45.18 12.50
Migration background, ref. native German	0.32 0.58
University degree, ref. no university degree	0.15 0.36
Neighbourhood characteristics: Share of . . .	
Welfare receivers	0.18 0.10
Unemployed	0.15 0.08
Long-term unemployed	0.11 0.06
Further spatial information: Lives in a . . .	
High-unemployment county	0.49 0.50
High-inequality neighbourhood	0.34 0.47
Urban county	0.51 0.50
Observations	1841

Notes. The table presents the mean and standard deviations for the main dependent, explanatory and core control variables. The table is based on misum and stems from a multiply-imputed data set.

in each grid cell. The lagged instead of same-year grid information ensured the correct temporal relation and reduced sorting bias.³

Main dependent and independent variables

The dependent variable was the stigma consciousness of the unemployed. The variable measures stigma consciousness with a normalised and unweighted sum of eight statements following the operationalisation of stigma consciousness by Linden et al. (2018) and Lang and Gross (2019) with a theoretical range of values from 0 to 100. The scale

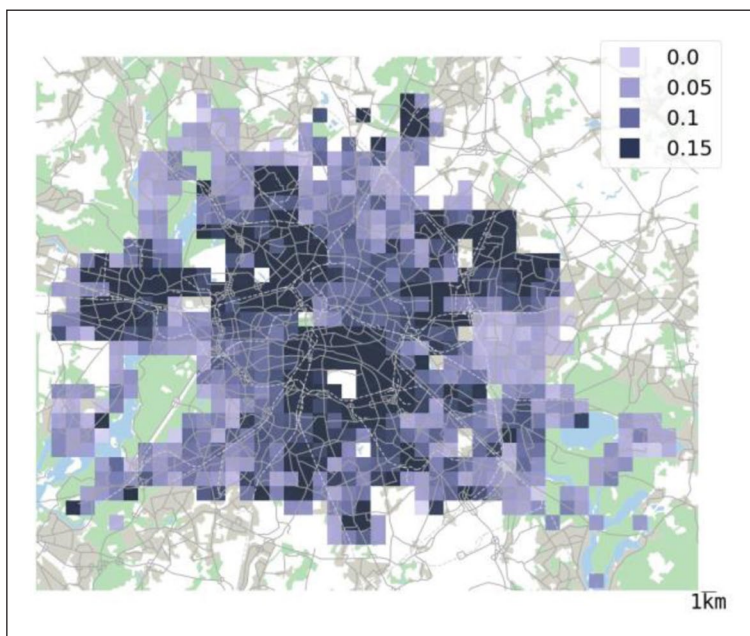


Figure 1. Distribution of neighbourhood unemployment in Berlin in 2012.

Notes: Each 1×1 km grid cell is one neighbourhood. Darker shades of purple indicate higher levels of unemployment. Blue areas in the background represent water; green areas, forests; light yellow areas, settlements; and solid grey lines, roads. The legend indicates the inclusive lower bound of each category, for example, 0.15 represents 15% unemployed or more.

was developed based on the Stigma Consciousness Questionnaire (Pinel, 1999) by Gurr and Jungbauer-Gans (2013: 343). It comprises statements such as ‘I feel burdened because of being unemployed’.⁴ All unemployed respondents rate each statement on a four-point Likert scale. The mean stigma consciousness was equal to 38.4 with a standard deviation of 13.6 (see Table 1). As not all unemployed answered all statements, 5.9% of the analytical sample showed missing values. Missing values are more often among second-generation migrants and respondents with three or more previous unemployment episodes (see Online Appendix Figure A.1).⁵

The main independent variable for all three hypotheses is the local prevalence of unemployment, measured as the share of unemployed in the respondents’ neighbourhood in the previous year. To calculate the share, we divided the number of all persons without any employment entry⁶ (henceforth: the unemployed) by the sum of all employed, unemployed, or persons in active labour market policy programmes in the respondents’ neighbourhood. Figure 1 provides an example of the high granularity of the neighbourhood data for Berlin. Darker shades of purple indicate higher neighbourhood unemployment rates and lighter shades lower rates. In our sample, the average unemployment share was equal to 15.2%.⁷

Unemployment and welfare receipt are highly correlated (sample correlation: 0.96). To disentangle these two concepts, the analyses used the share of unemployed and the

share of welfare receivers as two alternative specifications to test whether the stigma is mainly associated with receiving welfare benefits (Celhay et al., 2025; Horan and Austin, 1974; Mood, 2010; Rogers-Dillon, 1995). As the data stem from the social security system, all short-term and long-term unemployed in the data receive unemployment benefits. However, working individuals whose wages do not cover their daily expenses (German 'Aufstocker') can also receive housing or unemployment benefits. If receiving welfare benefits is the underlying mechanism in the relationship between neighbourhood unemployment and the individual's stigma consciousness, the estimate for the neighbourhood's share of welfare recipients should be larger and more precisely estimated than the estimate for the share of the unemployed. In the analysis sample, 17.8% of the respondents' neighbours received welfare benefits.

In adding the share of individuals not working for one year or more (long-term unemployed), the three definitions also serve as a test for gradual stages of local norm violations. Receiving welfare benefits while working could be considered a negligible norm deviation. Conversely, long-term unemployment should be the most severe and most visible deviation from a local employment norm. Hence, the last specification should report the strongest relationship with the individuals' stigma consciousness.

To test the nesting hypothesis, the article additionally employed official data on county unemployment rates from the Statistics Department of the Federal Employment Agency. These annual statistics report the share of unemployed individuals in the active workforce for each county and also originate from the social security data. In total, the analysis relied on unemployment statistics from 134 counties (13.73 respondents on average). The median unemployment rate in these 134 counties was equal to 8.6% in 2012. A binary variable indicated whether respondents live in counties below and above the median share of unemployed.⁸ Following the nesting hypothesis, respondents from counties with low unemployment should show stronger associations as the employment norm is intact and more often enforced.

For the coexistence hypothesis, an indicator of living in high-inequality neighbourhoods measured local coexistence of advantaged and disadvantaged residents within neighbourhoods. The used measure is the neighbourhoods' Gini coefficient⁹ from the grid data, where higher values indicate higher inner-neighbourhood inequality in terms of income, that is, resource and power endowment. While other segregation indices like dissimilarity indices require shares of categorical group memberships (like ethnicity shares), the Gini coefficient is especially suitable for income-based analyses on inequality (Pedersen, 2004). High-inequality neighbourhoods were grid cells from the highest tercile in the Gini coefficient distribution, whereas low-inequality neighbourhoods were grid cells from the lowest tercile in the Gini coefficient distribution.¹⁰

Controls

The models controlled for relevant individual-level mechanisms for explaining stigma consciousness discussed by Lang and Gross (2019): First, the *applicability of the norm* (gender, children in the household, care obligations, and age): they argue that the relevance of the employment norm and the stigmatisation potential varies with various socio-demographic factors such as gender or care obligations. Second, the *extent of the*

norm deviation (the duration of current unemployment and the number of unemployment episodes): this is essentially the same argument as for our different definitions of the extent of norm deviation in the neighbourhood, but on the individual level, where longer and repeated unemployment episodes pose a stronger deviation from the norm. Third, the *visibility of the deviation* (operationalised by a material deprivation index and the logarithmic household equivalent income): visibility is particularly high when unemployment and the associated financial restrictions manifest in material deprivation and cannot be compensated for by other household income. Fourth, the *formal societal control* (search obligation and welfare receipt): these are institutional forms of social control. Subjective health, education, and migration background were also included, because they are linked to stigma consciousness and to job market opportunities, such as re-employment probability (Lang and Gross, 2019). By controlling for residential moves, the models ensure that respondents have been exposed to a neighbourhood for a sufficient time.¹¹

On the neighbourhood level, the models controlled for the median wage and the number of residents to account for differences in the endowment and general socio-economic status between neighbourhoods. To account for further differences in the spatial-economic endowment, the models incorporated a dummy of whether the neighbourhood is in an urban or rural ('peripheral') region.¹² To control for sudden decreases in employment opportunities (demand shocks), the model controls for the number of firm closures in the neighbourhood in the last five years.

While this empirical set-up eliminated many potential confounders, it cannot claim causality. Unobserved heterogeneity between respondents or regions may bias the results. Random slopes (RS) models provide the best way to account for cluster-driven heteroscedasticity and cluster-correlated errors due to unobserved heterogeneity (Heisig and Schaeffer, 2019).

Methods

The article tests the hypotheses using linear multilevel random intercept (RI) regression models with RS for the unemployment rate.¹³ The unemployed represent level 1, the neighbourhoods level 2, and the counties level 3. In contrast to models with RI only, the specified RS models allow for differences in the association of stigma consciousness (level 1) and neighbourhood unemployment (level 2) between counties (level 3). The (simplified) model can be written as separate micro- (1), meso- (2), and macro-models (3)–(5). The complete model (6) can be transformed to separate the fixed and the random parts of the model (7) – with the random part in parentheses at the end of the equation:

$$stigma_{ijk} = \beta_{0jk} + X'_{ijk}\beta_1 + \varepsilon_{ijk} \quad (1)$$

$$\beta_{0jk} = \beta_{0k} + \beta_{2k}rate_{jk} + \beta_{3k}rate_{jk}^2 + M'_{jk}\beta_4 + v_{0jk} \quad (2)$$

$$\beta_{0k} = \beta_0 + Z'_k \beta_5 + \zeta_{0k} \quad (3)$$

$$\beta_{2k} = \beta_2 + \zeta_{2k} \quad (4)$$

$$\beta_{3k} = \beta_3 + \zeta_{3k} \quad (5)$$

$$\begin{aligned} stigma_{ijk} = & (\beta_0 + \zeta_{0k} + v_{0jk}) + X_{ijk}' \beta_1 \\ & + (\beta_2 + \zeta_{2k}) rate_{jk} + (\beta_3 + \zeta_{3k}) rate_{jk}^2 \\ & + M_{jk}' \beta_4 + Z_j' \beta_5 + \varepsilon_{ijk} \end{aligned} \quad (6)$$

$$\begin{aligned} stigma_{ijk} = & \beta_0 + X_{ijk}' \beta_1 + \beta_2 rate_{jk} + \beta_3 rate_{jk}^2 \\ & + M_{jk}' \beta_4 + Z_j' \beta_5 \\ & + (\zeta_{0k} + v_{0jk} + \zeta_{2k} rate_{jk} + \zeta_{3k} rate_{jk}^2 + \varepsilon_{ijk}) \end{aligned} \quad (7)$$

Here, $stigma_{ijk}$ is the dependent variable, the stigma consciousness of the unemployed i in neighbourhood j in county k . The explanatory variable, the neighbourhood unemployment rate in neighbourhood j in county k in the year before the survey, is denoted by $rate_{jk}$. The indices jk indicate that $rate_{jk}$ is a level-2 variable and varies on the neighbourhood level. To align the empirical set-up as closely as possible to H1, the model includes $rate_{jk}$ in quadratic terms to allow for U-shaped associations between the explanatory and the outcome variable. The RI is included in the model by the error terms ζ_{0k} and v_{0jk} , the RS by the error terms ζ_{2k} and ζ_{3k} . Finally, further controls on the individual X_{ijk} , neighbourhood M_{jk} , and the county level Z_j are included. ε_{ijk} denotes the individual-level error term.

To test H2 and H3, two separate models include interactions between the spatial moderators and the neighbourhood unemployment rate. For H2, the RS models incorporate cross-level interactions to test for differences in the slopes for the unemployment rate between counties. For H3, unemployment rate and Gini index are both on the neighbourhood level.

As usual for survey data and as already described for the administrative data, at least three sources of missing values are prevalent: (1) item non-response, (2) censored neighbourhood data, and (3) missing consent to linkage between the survey and the administrative data.

To deal with the sources of missing values (1) and (2) *multiple imputation by chained equations* (MICE; White et al., 2011; van Buuren, 2012) was applied. While MICE is not feasible for (3), missing consent was the case for only 0.5% of our sample. Moreover, Antoni and Bethmann (2019) report only a very small consent and linkage bias. MICE is particularly suitable because it can handle different types of variables and complex missing-data patterns flexibly. Each variable can be imputed using a model appropriate to its scale, which allows the procedure to capture the heterogeneous structure of survey and administrative data more realistically. Moreover, MICE can accommodate arbitrary

Table 2. Multilevel results of neighbourhood unemployment on stigma consciousness of the unemployed.

	Null model (1)	Share of . . .		
		welfare receivers (2)	unemployed (3)	long-term unemployed (4)
Base coefficient		−21.33 (−1.77)	−33.03* (−2.33)	−44.50* (−2.28)
Squared term		38.81 (1.54)	74.72* (2.30)	150.3* (2.25)
Constant	38.39*** (89.82)	43.44*** (11.23)	44.11*** (11.42)	43.42*** (11.67)
Var RI grid υ_{0jk}	0.971*** (4.17)	−9.903 (−0.05)	−10.31 (−0.05)	−10.17 (−0.06)
Var RI county ζ_{0k}	1.487*** (7.79)	1.383*** (6.60)	1.384*** (6.64)	1.382*** (6.61)
Error variance ε_{ijk}	2.528*** (95.61)	2.475*** (93.22)	2.475*** (93.34)	2.475*** (93.42)
Var RS 1 ζ_{2k}		−10.21 (−0.04)	−10.80 (−0.05)	−7.837 (−0.05)
Var RS 2 ζ_{3k}		0.734* (2.16)	0.694 (1.85)	0.691 (1.82)
Number of persons	1841	1841	1841	1841

Notes. The models include RI for grids and counties as well as RS for the neighbourhood unemployment (columns 2–4). The dependent variable is the stigma consciousness of the unemployed, ranging from 0 to 100. Higher values indicate higher stigma consciousness. T-statistics in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Columns 2–4 control for characteristics on the individual level (gender, presence of children, care obligations, age, unemployment duration and episodes, job search activity, social benefit transfers, subjective health, migration background, education and household relocations in the previous wave), the household level (level of deprivation, household income), the neighbourhood level (total number of residents, mean wage, number of firm closures in neighbourhood), and on the county level (urban or rural county).

missing-data patterns (assuming Missing at Random) and iteratively accounts for inter-dependencies between variables. For the analyses, $i = 50$ imputations using $m = 100$ iterations each were generated.

Results

Neighbourhood unemployment and stigma consciousness

Table 2 reports the null (restricted) model and three unrestricted models using one of the three different measures of the employment norm: the neighbourhood’s share of welfare recipients (column 2), unemployed (column 3), and long-term unemployed (column 4).

The null model shows that the RI models rely on substantial variation of stigma consciousness between neighbourhoods and counties. While all unemployment-related

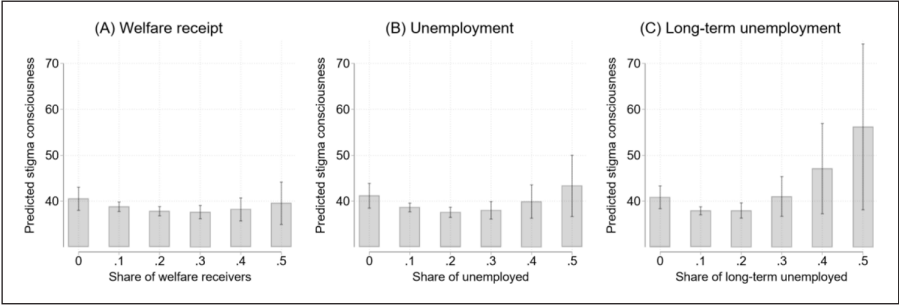


Figure 2. Visualising the stigma consciousness of the unemployed across the distribution of neighbourhood unemployment: predicted stigma consciousness values.
Notes. The dependent variable is the stigma consciousness of the respondent. 95% CI displayed. Controls mirror those in Table 2.

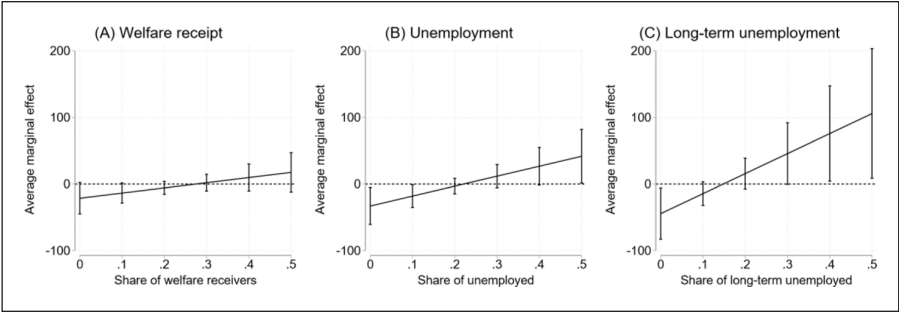


Figure 3. Marginal effects for neighbourhood unemployment on stigma consciousness of the unemployed.
Notes. The dependent variable is the stigma consciousness of the respondent. 95% CI displayed. Controls mirror those in Table 2.

variables in Table 2 (columns 2 to 4) show a negative and significant base coefficient and a positive and significant quadratic term, all coefficients increase in absolute size with increasing local intensities of norm-violating behaviour. To check for the proposed U-shaped association with neighbourhood unemployment proposed in H1, Figure 2 shows the predicted stigma consciousness across the distribution of the share of welfare receivers, unemployed and long-term unemployed (Panel A to C).¹⁴

In Figure 2, the U-shaped association between neighbourhood unemployment (x-axis) and the predicted stigma-consciousness (y-axis) becomes directly evident. Compared to neighbourhoods with medium shares of unemployment, the stigma consciousness of the unemployed is higher in neighbourhoods with low and high unemployment shares. To evaluate whether these differences are significant, Figure 3 reports marginal effects. For unemployment rates up to about 10%, Panel B in Figure 3 shows negative and significant associations between higher neighbourhood unemployment and the respondent's stigma consciousness. Up to this threshold, the stigma consciousness of the unemployed

decreases. In line with H1, Panel B reports a positive association between neighbourhood unemployment and the respondent's stigma consciousness for unemployment shares above 30%.¹⁵ Supplementary analyses on individual experiences of prejudice (Online Appendix Figure A.2) do not point towards experienced informal sanctions being the sole driver of this association.

Standardised regressions compare the coefficient sizes to other factors identified in the literature (see Online Appendix Table A.3). When comparing standardised values, the magnitude of the association from all unemployment measures (for all measures around -0.06) roughly relates to the influence of having children aged between 4 and 9 years (-0.065) on the individuals' stigma consciousness and but is smaller than the association with material deprivation (0.230).

When comparing three definitions of norm violations, Panel A in Figure 3 shows the smallest coefficients over the distribution and does not report significant estimates for high shares of welfare recipients in the neighbourhood. For the most severe norm-violation, Panel C reports significantly positive coefficients already at the threshold of 30%. Negative coefficients are significant below the 10% threshold. The finding of increasing coefficient sizes with increasing severity of norm-deviating behaviour suggests that the employment norm drives the increasing stigma consciousness of the individual and not the stigmatisation of welfare receipt.

The findings partly support previous international research (e.g. Celhay et al., 2025; Clark, 2003) and the predictions of social contagion theory. Up to a threshold of 10% of local unemployment, the results support the hypothesis that increasing unemployment weakens the employment norm and reduces the stigma consciousness of the unemployed. Above a share of 30% respectively 40% unemployed, the results of a higher stigma consciousness with higher local unemployment contradict previous research but confirm H1.

Spatial moderators

Figure 4 shows the marginal effects for the test on nested spatial structures (H2). The figure displays the interaction of neighbourhood unemployment and living in counties above (high-unemployment) and below (low-unemployment) the sample median. For respondents living in low-unemployment counties (black estimates), neighbourhood unemployment and stigma consciousness are positively and significantly related. For respondents living in high-unemployment counties (grey estimates), no panel provides significant effects across the neighbourhood's unemployment distribution. Hence, the results support H2, which predicted that the unemployed feel more stigmatised in high-unemployment neighbourhoods located in counties with few other unemployed. The norm deviation of the unemployed in such counties is more visible – not only for potential 'stigmatisers' but also for the individuals themselves.

Figure 5 shows the marginal effects for the test on the relevance of socio-economic coexistence (H3). The figure displays the interaction of neighbourhood unemployment and living in neighbourhoods on the upper (high-inequality) and the bottom (low-inequality) tercile of the Gini sample distribution. Despite large CI for low-inequality neighbourhoods, Figure 5 reveals that living in proximity with affluent individuals amplifies the psychological salience of norm deviations and the internalisation of stigma, that is in

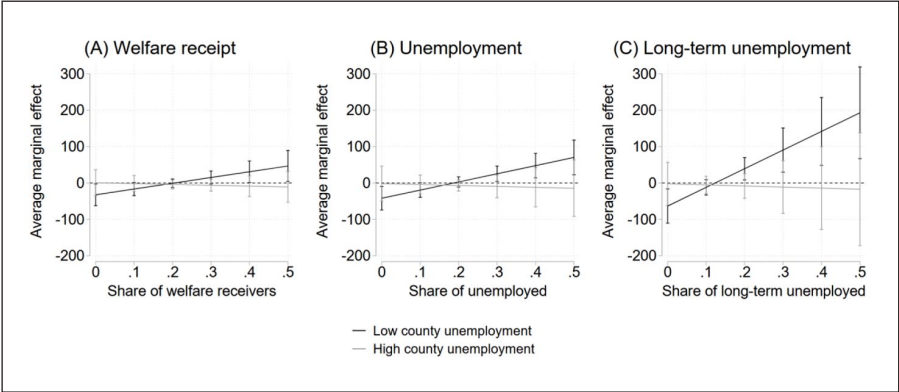


Figure 4. Testing the mechanism of nested spatial units: marginal effects for neighbourhood unemployment on stigma consciousness of the unemployed by county unemployment.
Notes. Low-unemployment counties are defined as those with an unemployment rate below 8.6% (sample median), high-unemployment counties as those with a share of 8.6% or higher. The dependent variable is the stigma consciousness. 95% CI displayed. Controls mirror those in Table 2.

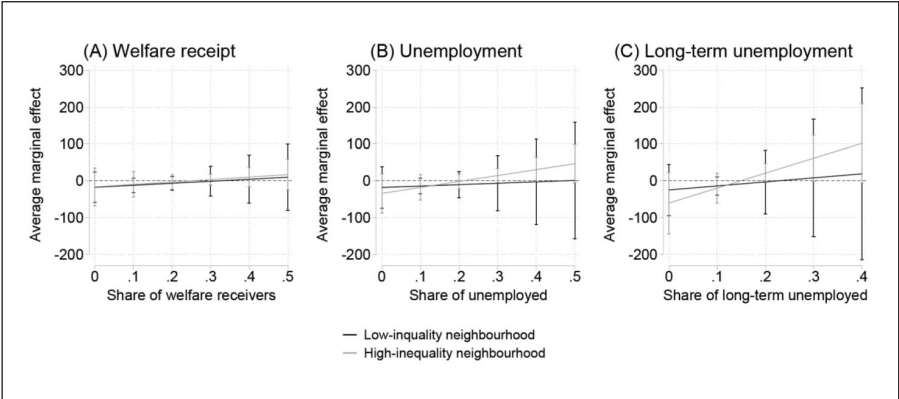


Figure 5. Testing the mechanism of socio-economic coexistence: marginal effects for neighbourhood unemployment on stigma consciousness of the unemployed by neighbourhood inequality.
Notes. Neighbourhood inequality is defined by income inequality measured with the Gini coefficient. Highest Ginitercile vs lowest Gini tercile inequality. The dependent variable is the stigma consciousness. 95% CI displayed. Panel C only displays estimations up to a share of 40% due to a lack of neighbourhoods exceeding this threshold for long-term unemployment. Controls mirror those in Table 2.

high-inequality neighbourhoods the stigma consciousness of the unemployed increases with higher neighbourhood unemployment. Contrarily, the relationship between neighbourhood unemployment and respondents' stigma consciousness for low-inequality neighbourhoods is not significant. Hence, the findings support H3.

Robustness analyses on neighbourhood size, urbanity or region (East vs West Germany) do not reveal similarly distinct patterns (see Panels A, B, and C in Online Appendix Figure A.5). Hence, the results in the Section *Neighbourhood unemployment and stigma consciousness* are likely to be driven by neighbourhoods with high inequality, or by neighbourhoods in otherwise low-unemployment regions. This finding suggests stigmatisation processes that are conditional on a sufficient number of *potentially stigmatised* persons, that is, the unemployed and other low-income residents and *stigmatising* persons, that is, high-income residents.

While sample size restrictions prevent further analyses on the intersection of living in high-inequality neighbourhoods from low-unemployment counties, the two subgroup analyses point towards two factors being relevant for explaining the spatial component in the unemployed's stigma consciousness: Not only is the existence of a local norm relevant, but also the prevalence of local norm enforcers.

Conclusion

Being unemployed is considered a stigmatised characteristic in many modern societies. This article uses the labelling approach and social contagion theory to derive theory-driven hypotheses describing the stigmatisation process. The article discusses the neighbourhood as one potential geographical level for norm enforcement. While social contagion theory only predicts lower stigma consciousness with increasing prevalence of local deviant behaviour, that is, neighbourhood unemployment, the article complements social contagion theory with the labelling approach to predict a U-shaped association between neighbourhood unemployment and individuals' stigma consciousness. To test the proposed hypotheses, the article combined rich survey data with georeferenced population data from the German Federal Employment Agency.

The findings support the hypothesis of a U-shaped association between neighbourhood unemployment and individuals' stigma consciousness. Unemployed residents in neighbourhoods with low- or very high-unemployment shares show the highest stigma consciousness. The intensity of this link strengthens with stricter definitions of norm-deviating behaviour in the neighbourhood.

Further results show that the composition of local social environments shapes the enforcement of employment norms and that some communities enforce them more rigorously than others. The multilevel models report the strongest positive associations between neighbourhood unemployment and individuals' stigma consciousness in low-unemployment counties and high-inequality neighbourhoods. Hereby, the findings for low-unemployment counties confirm the relevance of nested spatial structures and, therefore, support previous literature on neighbourhood stigmatisation as an additional source of emotional distress and disadvantage (Besbris et al., 2015; Wacquant, 2008). Similarly, the findings confirm the relevance of socio-economic coexistence in highlighting the role of present stigmatisers for the sanctioning of deviations from global employment norms. In both analyses, stronger – or more visible – signals from other (deviating) individuals predict higher individuals' stigma consciousness.

Although the article strives for the best possible identification and modelling, it also has limitations. First, the article does not claim causality. The only data source to study

the research question at hand is the cross-sectional survey data. This snapshot only insufficiently captures the inherently dynamic process of norm-setting, deviation, and enforcement. Sorting bias due to dynamic adaptation processes like residential mobility and endogenous neighbourhood change limits the article's scope, especially if more or less resilient respondents sort into high-unemployment neighbourhoods. Nevertheless, the final models try to control for all confounders, including residential mobility, and at least ensure the correct temporal order for causal identification concerning the neighbourhood unemployment.

Moreover, the measure of neighbourhoods using 1×1 km grid cells might still be too broad and likely results in measurement error, which could bias the results against zero (Fotheringham and Wong, 1991; Ostermann, 2026). However, survey data do not allow using spatial entities on a lower level in the multilevel model, because the relation between unemployed respondents and grid cells is already low (1.6 persons per grid on average).

Lastly, the article cannot analyse how the local employment norm is enforced: Do the unemployed already feel more stigmatised in observing other more affluent residents or do they feel more stigmatised because they experience informal sanctions by others? While supplementary analyses on individual experiences of stigmatisation point more towards internalisation of social identity and observation, the analyses ultimately cannot dive deeper into the stigmatisation process. Gurr and Lang (2018) give some qualitative insights on questions like this in general, but direct quantitative tests on possible mechanisms would be desirable for the future.

In the context of existing research, the results point towards several general policy implications. The article's findings suggest that higher local unemployment may lead (in specific contexts) to higher stigma consciousness. This may lead to a vicious circle: higher stigma consciousness leads to negative individual outcomes such as subjective well-being, health, and re-employment expectations and realisations (Krug et al., 2019). Other policy implications are more complex and come with several counterbalancing effects. If policymakers wish to reduce the emotional distress of the unemployed, the results of the studies imply that relocating such individuals to neighbourhoods with lower levels of unemployment would be one way to do so – which is neither feasible nor ethically justifiable. Nevertheless, targeted information about prevalent local norms and objective reflection about social forces at work might be a way to buffer the negative effects of informal social control on the unemployed's stigma consciousness. Leveraging one identified mechanism of the article, such targeted interventions could focus on neighbourhood stigmatisation. First studies (e.g., Schwarze, 2022) already show that local media contribute to neighbourhood stigmatisation.

For Germany specifically, these findings suggest that activation policies under SGB II should be regionally differentiated rather than uniformly designed: in low-unemployment counties with high-unemployment neighbourhoods, Jobcenter interventions should explicitly address the heightened stigma consciousness of clients, for instance, through stigma-aware counselling embedded in existing case management. Beyond individual-level measures, the 2023 Bürgergeld reform underscores that symbolic policy changes only translate into reduced stigmatisation if they are reinforced by local discourse, which calls for closer cooperation between regional employment agencies and municipal communication actors to counter territorial stigmatisation.

The findings also allow important theoretical considerations. The analyses show that not only a sufficient local number of potentially *stigmatised* persons is necessary but also of *stigmatising* persons. The article points toward inequality within and between neighbourhoods as the main driver of this association. Although the results report a negative effect of local unemployment on the individual's stigma consciousness (as the labelling approach and social contagion theory predict), this effect is offset and even reversed by the effect of being a member of an increasingly larger stigmatised group.

Moreover, the article contributes to the literature on employment norms in general by shedding light on the relevance of different spatial levels. The results suggest different relevant spatial levels for norm setting (county level or higher) and norm enforcement (neighbourhood level). Norm enforcement requires direct action in terms of individual efficacy, that is, individuals' willingness to intervene (Horne and Mollborn, 2020; Sampson and Groves, 1989). The results imply that narrower community measures capture such mechanisms better. In contrast to norm enforcement, norm formation and setting require long-term and subsequent social processes that do not necessarily require interaction and are passed (un)consciously from one generation to the next (Schultner et al., 2024). However, setting a norm may need more individuals conforming to the norm than just enforcing it, making higher spatial levels than the neighbourhood more likely to measure such norms appropriately. This highlights the importance of very detailed arguments on a theoretical level, but also when deciding about identification or modelling strategies.

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Data availability statements

Data access for the survey data (PASS) can be provided via a Scientific Use File supplied by the Research Data Centre (FDZ) of the German Federal Employment Agency (BA) at the Institute for

Employment Research (IAB). The administrative data (GridAB) analysed during the current study are not publicly available as the authors use administrative data of the IAB. The data are social data with administrative origin, which are processed and kept according to Social Code III. There are certain legal restrictions due to the protection of data privacy. The data contain sensitive information and therefore are subject to the confidentiality regulations of the German Social Code (Book I, Section 35, Paragraph 1). The data are held by the IAB (email: iab@iab.de, phone: +49 911 1790) and are onsite available on reasonable request. The authors are willing to assist replications or reproductions (Kerstin.ostermann@uni-bielefeld.de).

Supplemental material

Supplemental material for this article is available online.

Notes

1. §2(1) SGB II states, ‘Persons capable of earning and entitled to benefits . . . must exhaust all possibilities to end or reduce their need for assistance. An employable person entitled to benefits must actively participate in all measures for their integration into work . . . Within the framework of priority self-help and personal responsibility, persons capable of earning and entitled to benefits should use their own potential and make use of the services of other providers.’
2. Among others, Fowler (2016) and Ostermann (2026) stress that neighbourhoods are dynamic entities with a more fluid, multi-scalar nature. Nested hierarchical structures can serve only as a proxy for such dynamics.
3. Using contemporaneous information only led to slightly more efficient estimations.
4. The list of all employed items (answer options: ‘fully applies’, ‘applies to a large extent’, ‘applies to a small extent’, ‘does not apply at all’):
 1. ‘It’s hard for me to keep relationships with employed people alive.’
 2. ‘I feel burdened because of being unemployed.’
 3. ‘There are situations in everyday life in which I become aware that life is more difficult for someone unemployed than for employed people.’
 4. ‘I think that most people have more prejudices against the unemployed than they would openly admit.’
 5. ‘I feel more obliged to other unemployed people than to employed people.’
 6. ‘Prejudices against the unemployed affect me personally.’
 7. ‘In certain situations I try to hide that I am unemployed.’
 8. ‘I try to avoid situations where prejudice or discrimination against the unemployed could occur.’
5. MICE increased the computational power and corrected for possible non-response biases, see Online Appendix Section 1.
6. In Germany, registered unemployed can receive the full amount of eligible unemployment benefits while working on a marginal basis. Our unemployment definition excludes those individuals.
7. With 15.2%, the sample average was higher than for the unmerged grids (6.3%; see Online Appendix Table A.1), indicating neighbourhood clustering of the unemployed.
8. Online Appendix Figure A.3 also runs the analyses for the surrounding 1 × 1 km grids, 33 local labour markets (LLM), and 16 national states while keeping the unemployment threshold of 8.5% the same, using LLM yields similar results compared to the main specification.
9. The sample Gini ranged between 0.29 and 0.67.

10. A variable splitting the sample at the median yielded similar results but with larger CI due to more noise (see Online Appendix Figure A.4).
11. Panel D in Online Appendix Figure A.5 interacted the neighbourhood unemployment with residential moves. These models find only significant associations for non-movers ('stayers').
12. The binary indicator subsumed 'central' and 'very central' regions to urban and 'peripheral' and 'very peripheral' regions to rural. The original classification stems from the Federal Institute for Research on Building, Urban Affairs and Spatial Development. For further information, see https://www.bbsr.bund.de/BBSR/DE/forschung/raumbeobachtung/Raumabgrenzungen/deutschland/gemeinden/Raumtypen2010_vbg/Raumtypen2010_LageSied.html.
13. Online Appendix Table A.2 provides results for RI models only. Although showing slightly larger coefficients and CI, the results remain the same.
14. Mimrgns reports 95% CI. However, these are based on an incorrect number of degrees of freedom. In practice, the difference between the correct and reported CI is small.
15. In the sample, 5.1% of respondents live in grid cells with a share of unemployed of 30% or higher.

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